PROJECT REPORT

Airline Passenger Satisfaction Analysis

A Power BI study of passenger feedback and flight delays

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Project area	Airline / Travel Analytics
Data preparation	Python – Google Colab
Dashboard tool	Microsoft Power BI

Submitted for project evaluation

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Project at a glance

I used the Airline Passenger Satisfaction dataset to look at satisfaction levels, service ratings, passenger groups and flight delays. I prepared the data in Python and built a four-page interactive report in Power BI. The work is descriptive: it explains patterns in the available data rather than predicting future passenger behaviour.

1. Introduction

For this project, I wanted to understand what the passenger feedback in an airline satisfaction dataset could tell us about the travel experience. The data includes passenger details, ratings for different services, flight distance and delay information. I used these fields to compare satisfaction across groups and to see which service areas may need attention.

The final output is a Power BI dashboard with four pages. It starts with an overall summary, then looks more closely at service ratings and passenger groups, examines flight delays, and finishes with business observations and limitations.

2. Problem Statement and Objectives

Airlines receive feedback from many passengers, but a large set of responses can be difficult to review one record at a time. The purpose of this analysis is to make the feedback easier to understand and to highlight areas that a business team could investigate further.

The project focuses on these questions:

- How many passengers in the dataset are satisfied and how many are not?
- Do satisfaction levels differ by travel class, customer type, travel purpose, age group or gender?
- Which service ratings show the largest difference between satisfied and not-satisfied passengers?
- How do satisfaction rates compare across departure- and arrival-delay groups?

The aim was to answer these questions with clear comparisons and an interactive report, not to claim that the analysis proves why every passenger felt a certain way.

3. Dataset and Preparation

I downloaded the Airline Passenger Satisfaction dataset from Kaggle. It contains 129,880 passenger records and 22 original columns. The data is structured in rows and columns, with one record representing a passenger response.

Some of the fields used in the analysis were:

- Passenger details: Gender, Age and Customer Type.
- Trip details: Type of Travel, Class and Flight Distance.
- Delay information: Departure Delay in Minutes and Arrival Delay in Minutes.
- Service ratings: Online boarding, Inflight entertainment, Seat comfort, On-board service, Leg room service and other service fields.
- Outcome: Satisfaction, recorded as satisfied or not satisfied.

Preparation steps

I opened the CSV dataset in Google Colab and checked its shape, column names, data types, missing values and duplicate records. I also reviewed the distribution of the satisfaction field before building the report.

There were 393 missing values in Arrival Delay in Minutes. I left these as missing rather than changing them to zero. A zero means that no delay was recorded, while a blank means that the arrival-delay value was not available. Treating the two as the same could affect the comparisons.

For the Power BI report, I created grouped fields such as Age Group, Flight Distance Group, Departure Delay Group and Arrival Delay Group. I also used a Satisfaction Label to make the two satisfaction categories easier to display in charts.

4. Tools Used

Tool	How I used it
Python / Google Colab	To inspect the dataset, check data quality and carry out initial preparation and analysis.
Microsoft Power BI	To build the data model, create DAX calculations, add visuals and slicers, and assemble the four report pages.
CSV	The file format used to bring the dataset into the analysis workflow.

5. Dashboard Design

I arranged the Power BI report into four pages so that a user can move from the overall picture to more detailed comparisons. The pages share the same colour theme and filter area. Page navigation buttons make it possible to move between the sections without relying only on the tabs at the bottom of Power BI.

Dashboard Page 1 – Satisfaction Overview

This page gives the main passenger totals and satisfaction percentages. It includes KPI cards for total passengers, satisfied passengers, not-satisfied passengers and average service rating. The charts compare satisfaction by travel class, customer type and type of travel. Slicers let the user narrow the results by passenger and trip attributes.

Dashboard Page 2 – Service & Passenger Analysis

This page compares service ratings for satisfied and not-satisfied passengers. It also shows satisfaction by age group, flight-distance group and gender. The aim is to make it easier to spot differences between service areas and passenger segments.

Dashboard Page 3 – Flight Delay Analysis

This page looks at departure and arrival delays. The delay-group visuals show passenger counts alongside satisfaction rates, using a secondary axis for the percentage line. Two additional charts compare average arrival and departure delays between the satisfaction groups.

Dashboard Page 4 – Business Insights & Recommendations

The final page brings together the main observations, suggested business actions and the limitations of the dataset. It is intended as a short summary for someone who wants the main message without reviewing every chart.

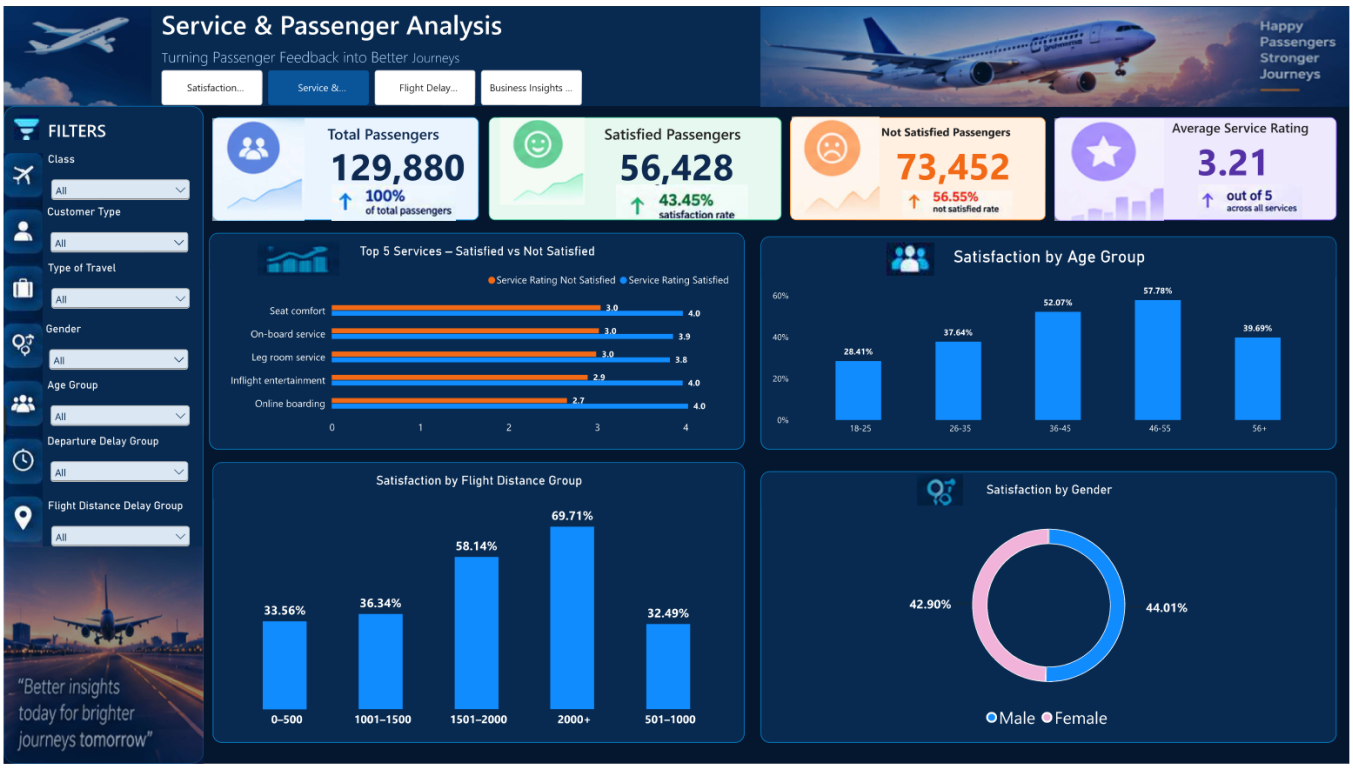
Dashboard Screenshot 1 – Satisfaction Overview

The first page presents the headline passenger measures and satisfaction comparisons by travel class, customer type and type of travel.



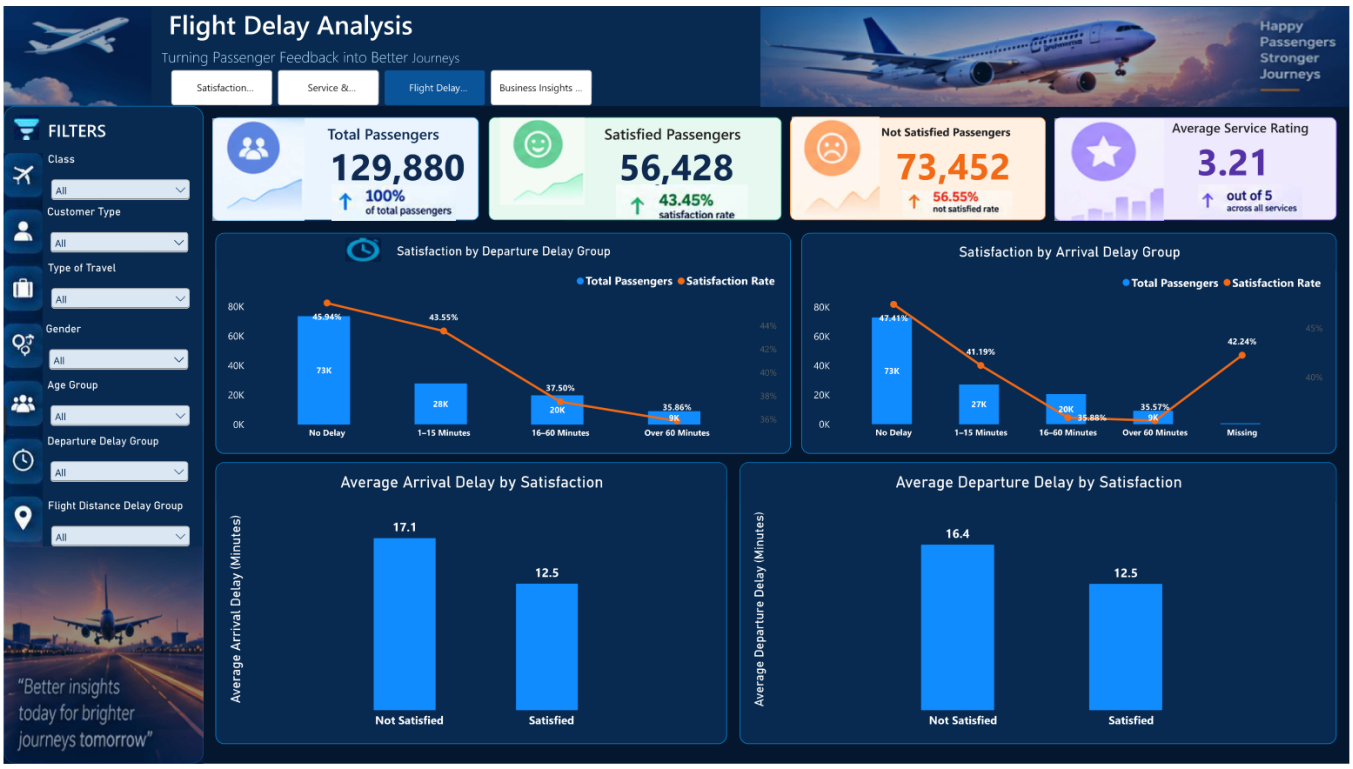
Dashboard Screenshot 2 – Service & Passenger Analysis

The second page compares service ratings and looks at satisfaction across passenger groups.



Dashboard Screenshot 3 – Flight Delay Analysis

The third page compares satisfaction across delay groups and shows average delays by satisfaction category.



6. Findings and Business Suggestions

The dashboard shows 129,880 passengers in total. Of these, 56,428 are marked satisfied (43.45%) and 73,452 are marked not satisfied (56.55%). The average service rating shown on the report is 3.21 out of 5.

Travel class: Business Class has the highest satisfaction rate in the report at about 69.15%. Economy Plus is about 24.53%, while Economy is about 18.97%. This is a difference worth exploring further with more detailed customer and operational information.

Service ratings: Online boarding, inflight entertainment and seat comfort show some of the largest rating gaps between satisfied and not-satisfied passengers among the services compared. These are sensible areas for the airline to review first.

Departure delays: The satisfaction rate shown for the no-delay group is 45.94%. It falls to 43.55% for delays of 1–15 minutes, 37.50% for delays of 16–60 minutes, and 35.86% for delays over 60 minutes. This pattern suggests that longer delays are associated with lower satisfaction in this dataset.

Average delays: The dashboard shows an average arrival delay of around 17.1 minutes for not-satisfied passengers and 12.5 minutes for satisfied passengers. The average departure delay is around 16.4 minutes and 12.5 minutes respectively.

Business suggestions: The airline could review the online boarding process, look at passenger feedback on inflight entertainment and seat comfort, and review how delay information is communicated to passengers. The passenger-group charts can also help teams decide where to investigate further.

These results describe patterns in this dataset. They do not show that a delay or a particular service rating by itself caused a passenger to be dissatisfied.

7. Challenges and Learning

- One challenge was handling the 393 blank arrival-delay values. I kept them blank rather than treating them as zero.
- I had to create useful categories for age, flight distance and delay duration so the comparisons would be easier to read.
- I spent time checking DAX calculations and making sure the measures responded to the categories and slicers in the visuals.
- Building the delay charts required a secondary axis because passenger counts and satisfaction percentages use different scales.
- I also adjusted the layout, titles, data labels, colours and navigation so the four pages looked consistent.

Working through these steps gave me practical experience in preparing data, writing DAX measures, choosing suitable visuals and explaining findings in a business context.

8. Limitations

- The dataset does not provide a suitable flight date or month field for the time-based analysis planned in this project.
- There are 393 missing values in Arrival Delay in Minutes. These records are kept as missing and are not treated as zero-delay flights.
- The analysis is descriptive. It shows relationships and differences in the available data, but it cannot prove that one factor caused dissatisfaction.
- The results describe the passengers included in this dataset. They may not represent every airline, route, airport or time period.
- Service ratings are based on passenger responses and may reflect individual expectations and experiences.

9. Conclusion and Future Scope

I completed a four-page Power BI dashboard using the airline passenger dataset. The report brings together overall satisfaction, service-rating comparisons, passenger segments and flight-delay analysis. It also includes a page of business observations and limitations.

The main result is that 56.55% of the passengers in this dataset are marked not satisfied. The comparisons also highlight rating gaps in some passenger services and lower satisfaction rates in the longer departure-delay groups. These findings give the airline areas to investigate, rather than a final explanation of every passenger's experience.

If the project is extended, I would add flight dates to study changes over time, refresh the analysis with more recent passenger records, and combine the survey data with route or operational information where available. A separate predictive model could also be developed in a future phase, with its performance tested properly.

10. References

Kaggle – Airline Passenger Satisfaction dataset (Binary Aviation Satisfaction 129K):
https://www.kaggle.com/datasets/arseniyshutko/binary-aviation-satisfaction-129k

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